APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Living vs Non-Living Organisms

SUBJECT: Environmental Studies (EVS101)

CLASS & SECTION: Class 2 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 49 / 50 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Draw and categorize 5 living and 5 non-living things found in school garden with biological criteria.

1. Executive Overview & Problem Context

All entities in our terrestrial environment are classified into living organisms (biotic) and non-living objects (abiotic). Living organisms possess the capacity for autonomous metabolic activity, respiration, growth, irritability (response to stimuli), reproduction, and cellular organization.

2. Core Analysis, Solutions & Systematic Working

- **The Seven Vital Life Processes (MRS GREN):**

  Organisms demonstrate: Movement (autonomous locomotion or phototropism), Respiration (gas exchange for ATP), Sensitivity (sensory response to temperature/light), Growth (cellular proliferation), Reproduction (continuation of species), Excretion (waste elimination), and Nutrition (intake of organic/inorganic sustenance).

- **Natural vs Artificial Non-Living Objects:**

  Non-living entities lack cellular structure and metabolic requirements. They are subdivided into Natural Abiotic Elements (sunlight, granite rocks, river water, atmospheric air) and Man-Made Artifacts (wooden benches, plastic watering cans, concrete pathways).

- **Garden Ecosystem Observations:**

  Observed Honeybee (Apis mellifera) foraging nectar; Earthworm aerating moist garden soil; Hibiscus rosa-sinensis photosynthesizing; Garden pebble remaining unchanged in mass and volume over time.

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3. Detailed Data Matrix & Systematic Findings

Subject Name	Classification	Can It Grow / Reproduce?	Key Observation
Marigold Plant	Living (Flora)	Yes (Seeds & Stems)	Leaves turn towards morning sunlight
Earthworm	Living (Fauna)	Yes (Lays cocoons)	Burrows into damp organic compost
Garden Water Hose	Non-Living (Synthetic)	No	Requires human manipulation to move
Granite River Rock	Non-Living (Natural)	No	Weathered by rain and physical friction
Caterpillar	Living (Fauna)	Yes (Metamorphosis)	Feeds voraciously on fresh mulberry leaves

4. Student Conclusions & Reflective Synthesis

Living organisms actively maintain homeostatic life cycles and depend upon non-living environmental elements (air, water, sunlight, soil) for sustained biological survival.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding biological observation! Clear understanding of life processes and environmental interaction."

APEX EVALUATION DIVISION

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